Research

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## What are microplastics?

Microplastics, MPs for short, are basically synthetic polymer particles smaller than 5mm in diameter. There are two types of MPs: primary and secondary. Primary MPs are MPs intentionally produced at a small size (like microbeads in skincare). Secondary MPs are the result of environmental fragmentation and weathering of larger plastic products (like plastic bags, bottles and fishing gear).

## Impact of microplastics on marine life

There are many impacts of MPs on marine life. There is widespread contamination of MPs in the ocean which leads to unintentional ingestion in aquatic animals. Marine fish are consistent consumers of microplastics, mistaking them for prey or ingesting them indirectly. Fish from areas with high human coastal population density and poor waste management (like the Asia-Pacific region) showed a higher prevalence and concentration of MPs. This points to land-based sources as a primary contributor. MPs accumulating in fish and other smaller marine life could lead to the harmful process of trophic transfer resulting in toxic bioaccumulation. This could lead to the endangerment and extinction of many marine species. These are indirect dangers to marine life. The direct dangers of MPs are when the plastics are already inside the body of the marine life (this is similar to MP impacts on humans). MP poisoning can cause obstructive damage and internal abrasions in the gastrointestinal tract. Furthermore, the accumulation of non-nutritive plastic can create a sense of **false satiety** (False satiety is a feeling of fullness that occurs without proper physiological reasons, often stemming from emotional triggers, the physical sensation of gas and bloating, or the consumption of liquids that don't provide long-lasting fullness.), reducing feeding and leading to energetic depletion and weight loss. The presence of foreign synthetic particles in the gut can trigger chronic inflammatory responses and suppress the immune system, making fish more susceptible to disease.

## Impact on human health

MPs are toxic, as they are not pure polymers, they can contain chemical additives (like phthalates, flame retardants and stabilizers) which can deeply affect the body on a physiological level. They can cause extreme damage to us through acting as a vessel for harmful pathogens. As the “Trojan Horse” effect states that MPs act as a vector for pathogens (Vibrio species). This could lead to greater damage to human health than the effect of the MP particle/contaminant alone.

## Scale of problem - stats

- We, as humans, ingest around 5 grams of plastic a week.
- A single wash and dry cycle can release over **700,000** microplastic fibers, with the amount depending on the fabric type, wash temperature, and machine type. For a 5kg load, estimates range from 600,000 to 17 million microplastic fibers released into the environment. Synthetic materials like acrylic release significantly more fibers than other fabrics.

**Major sources of microplastics - clothing, fibres etc**

- Aquatic environments
- Sewage
- Atmospheric particulates
- Living objects in a food chain (eg fish)
- Clothing and fibres

**How Refyber helps:**

Our organisation helps by preventing the microfibers released from clothing in the dryer to enter the ocean. In a usual household, there are filters that prevent the microplastics from entering the water stream however, these microplastics end up getting thrown into the trash which will ultimately end up in the ocean from landfill. Therefore, Refyber focuses on reducing this by repurposing the microplastics filtered out by the dryers and turning them into eco-paper. This allows a break from the cycle of harm and instead provides opportunity for education and awareness, whilst making a positive physical impact as well.

**Arabic gum**

Gum arabic (GA) is the oldest gum known for its diverse application for more than 5000 years. The gum is produced from some species of Acacia, mainly growing/planted in arid and semi-arid regions of Africa and Asia.

Sudan is the leading producer and exporter of GA in the world. GA is a mixture of potassium, magnesium and calcium salts of polysaccharide acid and is easily soluble in water.

https://www.tandfonline.com/doi/pdf/10.1080/10942912.2013.809541

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Gum arabic is not inherently "better" than resin; rather, they are fundamentally different substances with distinct properties and applications. The primary difference is that gum arabic is water-soluble, while resins are typically alcohol- or oil-soluble and not water-soluble, which dictates their specific uses and advantages in various contexts

## Key Differences and Advantages

| Feature | Gum Arabic (a gum) | Resin (e.g., pine resin) |
|---|---|---|
| **Solubility** | Highly water-soluble; insoluble in alcohol/oil | Alcohol/oil soluble; insoluble in water |
| **Composition** | Polysaccharides and glycoproteins | Complex mixtures, often terpenes and essential oils |
| **Toxicity/Edibility** | Edible, non-toxic, and biocompatible | Can be toxic or cause irritation; generally not edible |
| **Primary Use** | Emulsifier, stabilizer, thickener in food/pharma, binder for water-based paints/inks | Aromatherapy, varnishes, specific adhesives, incense |
| **Cleanup** | Easy to clean with water | Difficult to clean from surfaces |

BENEFITS OF GUM ARABIC

- Food and Pharmaceutical Applications: Gum arabic is widely used in human consumption products (beverages, candies, medications) because it is edible and non-toxic, unlike most resins.
- Aqueous Solutions/Emulsions: Its high water solubility makes it an excellent natural emulsifier and stabilizer for water-based systems (e.g., keeping oil and water mixed in salad dressings or soft drinks).
- Art Supplies: It is the standard, easy-to-use binder for watercolors, allowing for smooth flow and easy cleanup with water.
- Eco-friendliness: Gum arabic is biodegradable and derived from a renewable source, making it an environmentally sustainable option